WEEKLY TESTS AND FINAL EXAM

Eleven 10-question weekly tests, a 20-question final, and a full answer key with worked solutions at the back.

Closed book, calculator allowed. Weekly tests take about 40 minutes and the final about 90. Number answers within 2% count as correct (angles within 1°). Pass mark: 70%.

Mark your own paper from the key, or type your answers into the interactive hub to have them marked.

WEEK 1 TEST · 10 QUESTIONS · 40 MIN

Units, vectors and adding forces

NAME _____ DATE _____ SCORE ____ / 10

Q1 Which of these quantities is a vector?

(A) Mass

(B) Time

(C) Force

(D) Volume

Q2 A replacement pump has a mass of 250 kg. What is its weight?

Answer _____ kN

Q3 The equilibrant of a force system is:

(A) The largest force in the system

(B) Equal in size to the resultant, on the same line of action, but in the opposite direction

(C) The sum of the magnitudes of all the forces

(D) Always perpendicular to the resultant

Q4 0.005 MN is the same as:

(A) 5 N

(B) 50 N

(C) 5 kN

(D) 50 kN

Q5 Two tugs pull a ship. One pulls with 40 kN and the other with 30 kN, and the angle between their towlines is 50°. What is the magnitude of the resultant pull?

Answer _____ kN

Q6 For the same two tugs, what angle does the resultant make with the 40 kN towline?

Answer _____ °

Q7 A 12 kN force acts at 35° above the horizontal. What is its horizontal component?

Answer _____ kN

Q8 A 500 N force acts along a slope that rises 3 units for every 4 units of run (a 3-4-5 triangle). What is its vertical component?

Answer _____ N

Q9 Three forces act on a ring: 200 N at 0°, 300 N at 90° and 250 N at 225°, with angles measured anticlockwise from +x. What is the magnitude of the resultant?

Answer _____ N

Q10 Two forces of 30 N and 40 N act on a point. Which of these *cannot* be the magnitude of their resultant?

(A) 5 N

(B) 10 N

(C) 50 N

(D) 70 N

WEEK 2 TEST · 10 QUESTIONS · 40 MIN

Equilibrium of a particle and free-body diagrams

NAME _____ DATE _____ SCORE ____ / 10

Q1 A particle is in equilibrium when:

(A) It is not moving

(B) The resultant force on it is zero

(C) All the forces acting on it are equal

(D) It has no weight

Q2 A rope passes over a frictionless pulley. The tension on the two sides of the pulley is:

(A) Greater on the loaded side

(B) The same on both sides

(C) Half on each side

(D) Zero on the free side

Q3 A load hangs from two equal sling legs. As the included angle between the legs increases from 60° to 120°, the tension in each leg:

(A) Decreases

(B) Stays the same

(C) Increases, from about 0.58W to 1.0W

(D) Doubles from 0.5W to 1.0W

Q4 A 500 kg load hangs from two sling legs, each 30° from the vertical. What is the tension in each leg?

Answer _____ kN

Q5 The same 500 kg load is now rigged with each leg at 60° from the vertical. What is the tension in each leg?

Answer _____ kN

Q6 A 10 kN weight hangs from point A. Cable AB runs horizontally to a wall. Cable AC rises at 40° above the horizontal to the deckhead. What is the tension in AC?

Answer _____ kN

Q7 In the same setup, what is the tension in the horizontal cable AB?

Answer _____ kN

Q8 A spring with stiffness k = 2 kN/m has an unstretched length of 300 mm. It is stretched to 450 mm. What force does it exert?

Answer _____ N

Q9 A 600 N force acts from the origin toward the point (3, 4, 12) m. What is its z-component?

Answer _____ N

Q10 A force has direction angles α = 60° and β = 60°. Its third direction angle γ (acute) is:

(A) 30°

(B) 45°

(C) 60°

(D) 90°

WEEK 3 TEST · 10 QUESTIONS · 40 MIN

Moments of a force

NAME _____ DATE _____ SCORE ____ / 10

Q1 The SI unit of a moment is:

(A) N

(B) N/m

(C) N·m

(D) J/s

Q2 The moment of a force about a point is zero when:

(A) The force is very small

(B) The line of action of the force passes through the point

(C) The force is horizontal

(D) The force is perpendicular to the lever

Q3 A 250 N force is applied perpendicular to a spanner, 0.4 m from the bolt centre. What torque does the bolt get?

Answer _____ N·m

Q4 The same 250 N force is applied at 0.4 m, but at 60° to the spanner handle. What is the torque now?

Answer _____ N·m

Q5 A force F = (300 **i** − 200 **j**) N acts at the point (2, 3) m. Find its moment about the origin. Give a signed answer with anticlockwise positive.

Answer _____ N·m

Q6 A crane boom 10 m long is raised to 40° above the horizontal. A 5 kN load hangs from its tip. What moment does the load produce about the boom foot?

Answer _____ kN·m

Q7 Position vector r = (1 **i** + 2 **j**) m and force F = (50 **k**) N. What is the magnitude of the moment M = r × F?

Answer _____ N·m

Q8 Varignon's theorem states that:

(A) A force can slide along its line of action

(B) The moment of a force equals the sum of the moments of its components about the same point

(C) Every action has an equal and opposite reaction

(D) Couples can be moved anywhere

Q9 A lever has two forces on it: 200 N acting anticlockwise at 0.5 m from the pivot, and 150 N acting clockwise at 1.2 m. What is the resultant moment? Give a signed answer with anticlockwise positive.

Answer _____ N·m

Q10 The perpendicular distance used in M = Fd is measured from the point to:

(A) The point where the force is applied

(B) The line of action of the force

(C) The nearest support

(D) The centroid of the body

WEEK 4 TEST · 10 QUESTIONS · 40 MIN

Couples, equivalent systems and distributed loads

NAME _____ DATE _____ SCORE ____ / 10

Q1 A couple moment is called a 'free vector' because:

(A) It has no magnitude

(B) It has the same moment about every point, so it can be moved anywhere on the body

(C) It can only act about the centroid

(D) It is always zero

Q2 Two opposite 150 N forces act along parallel lines 0.25 m apart. What is the couple moment?

Answer _____ N·m

Q3 A 12 kN force acts 0.8 m from point O. You replace it with an equal force acting at O plus a couple. How big is the couple?

Answer _____ kN·m

Q4 A uniform load of 5 kN/m acts over 8 m. What is the equivalent resultant force?

Answer _____ kN

Q5 A triangular load rises from 0 at A to 9 kN/m at B, over 6 m. What is the resultant force?

Answer _____ kN

Q6 For the same triangular load, how far from A does the resultant act?

Answer _____ m

Q7 Three parallel downward loads act on a beam: 2 kN at x = 1 m, 5 kN at x = 3 m and 3 kN at x = 6 m. Where does their 10 kN resultant act?

Answer _____ m

Q8 A force can be moved along its own line of action without changing its external effect on a rigid body. This is the principle of:

(A) Moments

(B) Transmissibility

(C) Superposition

(D) Virtual work

Q9 A plate has a 40 N·m anticlockwise couple, a 25 N·m clockwise couple, and a couple made of two 100 N forces 0.3 m apart acting clockwise. What is the net couple moment? Give a signed answer with anticlockwise positive.

Answer _____ N·m

Q10 A trapezoidal load goes from 2 kN/m at A to 6 kN/m at B over 4 m. How far from A does its resultant act?

Answer _____ m

WEEK 5 TEST · 10 QUESTIONS · 40 MIN

Equilibrium of a rigid body

NAME _____ DATE _____ SCORE ____ / 10

Q1 How many unknown reactions does a fixed (built-in) support have in 2D?

(A) 1

(B) 2

(C) 3

(D) 4

Q2 A roller support provides:

(A) A force perpendicular to the surface it rolls on

(B) Horizontal and vertical forces

(C) A moment only

(D) A force parallel to the surface

Q3 A simply supported beam spans 5 m and carries a 20 kN load 2 m from A. What is the reaction at A?

Answer _____ kN

Q4 A 3 m cantilever is fixed at A. It carries a 4 kN load at the free end and a uniform load of 2 kN/m along its full length. What is the moment at the fixed support?

Answer _____ kN·m

Q5 For the same cantilever, what is the vertical reaction at A?

Answer _____ kN

Q6 A beam has supports at A (x = 0) and B (x = 4 m), and a 6 kN load at the end of a 2 m overhang (x = 6 m). What is the vertical reaction at A? Give a signed answer with up positive.

Answer _____ kN

Q7 A two-force member (no weight, loaded at two points only) carries forces that are:

(A) Perpendicular to the member

(B) Equal, opposite and along the line joining the two points

(C) Always in tension

(D) Unknown in direction until you solve

Q8 A 1.5 m hand-pump lever pivots 0.3 m from the pump rod end. A 200 N force is applied at the handle end. What force goes into the pump rod?

Answer _____ N

Q9 A horizontal beam AB, 4 m long, is pinned at A. At B it is held by a cable that makes 30° with the beam. A 10 kN load acts at mid-span. What is the cable tension?

Answer _____ kN

Q10 For the same beam, what is the horizontal pin reaction at A?

Answer _____ kN

WEEK 6 TEST · 10 QUESTIONS · 40 MIN

Trusses, frames and machines

NAME _____ DATE _____ SCORE ____ / 10

Q1 Simple truss analysis assumes that:

(A) Members are rigidly welded

(B) Members are joined by smooth pins and loads act only at joints, so each member carries only tension or compression

(C) Every member carries bending

(D) Joints can carry moments

Q2 At an unloaded joint where only two members meet, and they are not in line:

(A) Both members are zero-force members

(B) Both carry equal tension

(C) One is in tension and one in compression

(D) You need the method of sections

Q3 With the method of sections, the most unknown member forces you can generally cut and solve is:

(A) 1

(B) 2

(C) 3

(D) 6

Q4 A triangular truss spans 8 m. Its apex is 3 m above mid-span and carries a 12 kN load. What is the magnitude of the force in one sloping member?

Answer _____ kN

Q5 The sloping members of that truss are in:

(A) Tension

(B) Compression

(C) Zero force

(D) Bending

Q6 What is the force in the horizontal bottom tie of the same truss?

Answer _____ kN

Q7 A section is cut through a truss of depth 2 m. Take moments about the top joint above the cut bottom chord. The left reaction is 30 kN at 6 m from the joint, and a 20 kN load acts 3 m from it. What is the force in the bottom chord?

Answer _____ kN

Q8 A pair of pliers has its handles gripped with 100 N each, applied 120 mm from the pivot. The jaws grip 30 mm from the pivot. What force is applied at the jaws?

Answer _____ N

Q9 A bell crank ABC pivots at B. Arm BA is 200 mm long and horizontal, and arm BC is 100 mm long and vertical. A 50 N vertical force acts at A. What horizontal force at C holds it in equilibrium?

Answer _____ N

Q10 When you take a frame apart, the forces at a pin connecting two members are:

(A) Twice as large on the longer member

(B) Equal and opposite on the two members

(C) Always vertical

(D) Zero

WEEK 7 TEST · 10 QUESTIONS · 40 MIN

Internal forces: shear force and bending moment diagrams

NAME _____ DATE _____ SCORE ____ / 10

Q1 A simply supported beam spans 6 m with an 18 kN load at mid-span. What is the maximum bending moment?

Answer _____ kN·m

Q2 A simply supported beam spans 10 m and carries a uniform load of 3 kN/m. What is the maximum bending moment?

Answer _____ kN·m

Q3 A 2.5 m cantilever carries an 8 kN load at its free end. What is the magnitude of the maximum bending moment?

Answer _____ kN·m

Q4 A 4 m cantilever carries a uniform load of 1.5 kN/m. What is the maximum shear force?

Answer _____ kN

Q5 On a beam, the maximum bending moment occurs where:

(A) The shear force is largest

(B) The shear force is zero or changes sign

(C) The load is largest

(D) The supports are

Q6 The slope of the bending moment diagram at any point equals:

(A) The load intensity

(B) The shear force at that point

(C) The deflection

(D) Zero

Q7 Over a length of beam carrying a uniform load, the shear and moment diagrams are:

(A) Both constant

(B) Shear sloping straight line, moment parabola

(C) Shear parabola, moment straight line

(D) Both straight lines

Q8 A simply supported beam spans 8 m with a 24 kN point load 2 m from A. What is the maximum bending moment?

Answer _____ kN·m

Q9 For the same beam, what is the bending moment 5 m from A?

Answer _____ kN·m

Q10 A loaded ship has heavy cargo concentrated amidships, while most of its buoyancy is towards the ends. The hull is:

(A) Hogging, with the deck in tension

(B) Sagging, with the deck in compression

(C) In pure shear

(D) Unstressed

WEEK 8 TEST · 10 QUESTIONS · 40 MIN

Dry friction and wedges

NAME _____ DATE _____ SCORE ____ / 10

Q1 The friction force on a body at rest:

(A) Always equals $\mu_s N$

(B) Is whatever is needed for equilibrium, up to a maximum of $\mu_s N$

(C) Depends on the contact area

(D) Acts perpendicular to the surface

Q2 Two surfaces have $\mu_s = 0.4$ and are pressed together by a 500 N normal force. What is the maximum static friction force?

Answer _____ N

Q3 A block on a tilting plate starts to slide when the plate reaches 22°. What is μ_s?

Answer _____

Q4 A 50 kg crate sits on a level deck with $\mu_s = 0.35$. What horizontal force will just start it moving?

Answer _____ N

Q5 A 100 kg crate is pushed up a 15° incline at steady speed, with $\mu_k = 0.2$ and the force parallel to the incline. What force is needed?

Answer _____ N

Q6 How much work is done pushing that crate 5 m up the incline?

Answer _____ kJ

Q7 The angle of repose of a material is:

(A) Always 45°

(B) Equal to the angle of static friction, so $\tan\theta = \mu_s$

(C) Equal to the angle of kinetic friction

(D) Independent of the material

Q8 What force, parallel to a 30° incline, stops a 100 kg crate sliding down if $\mu_s = 0.25$?

Answer _____ N

Q9 A tall cabinet is pushed horizontally near its top. It will tip rather than slide if:

(A) The push needed to tip it is less than the push needed to slide it

(B) μ_s is very small

(C) It is very heavy

(D) The floor is level

Q10 A cabinet is 1.2 m tall and 0.6 m wide and weighs 800 N, with $\mu_s = 0.4$. It is pushed horizontally at its top edge. What push starts it moving, either by tipping or by sliding?

Answer _____ N

WEEK 9 TEST · 10 QUESTIONS · 40 MIN

Screws, belts, bearings and rolling resistance

NAME _____ DATE _____ SCORE ____ / 10

Q1 A rope is held stationary over a fixed drum with 180° of wrap and $\mu = 0.25$. The holding side has 500 N. What is the largest load the other side can carry without slipping?

Answer _____ N

Q2 A 15 kN load is to be held on a bollard with 150 N, with $\mu = 0.3$. How many turns of rope are needed? Give the exact, non-whole number.

Answer _____ turns

Q3 A belt drive has a tight-side tension of 900 N, $\mu = 0.3$ and 160° of wrap on the smaller pulley. What is the least slack-side tension before slip?

Answer _____ N

Q4 If that belt runs at 10 m/s, what power does it transmit?

Answer _____ kW

Q5 A square-threaded screw jack has a mean diameter of 40 mm, a lead of 8 mm and $\mu_s = 0.15$. What torque raises a 20 kN load?

Answer _____ N·m

Q6 A square-threaded screw is self-locking when:

(A) The lead angle θ is greater than the friction angle φ_s

(B) The friction angle φ_s is greater than the lead angle θ

(C) $\mu = 0$

(D) The lead equals the pitch

Q7 A 100 mm diameter shaft runs at 300 rev/min in a journal bearing with a radial load of 30 kN and $\mu = 0.05$. What power is lost to friction?

Answer _____ kW

Q8 A thrust collar has inner radius 50 mm and outer radius 100 mm. It carries an axial load of 10 kN with $\mu = 0.08$. What is the friction torque, assuming uniform pressure?

Answer _____ N·m

Q9 A 2000 kg machine is moved on rollers with radius 0.15 m. The coefficient of rolling resistance is $a = 0.5$ mm. What horizontal force keeps it rolling?

Answer _____ N

Q10 In the belt friction equation $T_2 = T_1 e^{\mu\beta}$:

(A) β is in degrees and $T_2 < T_1$

(B) β is in radians and T_2 is the larger tension

(C) β is in turns

(D) It only applies to V-belts

WEEK 10 TEST · 10 QUESTIONS · 40 MIN

Centre of gravity, centroids and fluid pressure

NAME _____ DATE _____ SCORE ____ / 10

Q1 A T-section is made of a web 20 mm wide by 180 mm tall (from y = 0 to 180 mm) with a flange 200 mm wide by 20 mm thick on top (y = 180 to 200 mm). Find the height $\bar{y}$ of its centroid above the bottom.

Answer _____ mm

Q2 A 300 mm × 200 mm plate has a 100 mm diameter hole centred 200 mm from its left edge. Find $\bar{x}$ of the plate, measured from the left edge.

Answer _____ mm

Q3 The centroid of a triangle lies:

(A) At mid-height

(B) One third of the height up from the base

(C) Two thirds of the height up from the base

(D) At the apex

Q4 A ship's lightship is 2000 t with KG 6.0 m. It loads 500 t of fuel at Kg 2.0 m and 300 t of cargo at Kg 8.0 m. What is the new KG?

Answer _____ m

Q5 What is the gauge pressure 6 m below the surface of seawater (ρ = 1025 kg/m³)?

Answer _____ kPa

Q6 A vertical door 1 m wide and 2 m high has its top edge at the surface of fresh water. What is the resultant water force on it?

Answer _____ kN

Q7 How far below the surface is the centre of pressure on that door?

Answer _____ m

Q8 A vertical hatch 1.2 m wide and 0.8 m high has its top edge 3 m below the surface of seawater. What is the resultant force on it?

Answer _____ kN

Q9 A ring has a circular cross-section of radius 20 mm, and its centroid is 200 mm from the axis. What is its volume (Pappus)?

Answer _____ litres

Q10 On a vertical submerged plane surface, the centre of pressure is:

(A) At the centroid

(B) Always below the centroid

(C) Always above the centroid

(D) At the free surface

WEEK 11 TEST · 10 QUESTIONS · 40 MIN

Moments of inertia and stability

NAME _____ DATE _____ SCORE ____ / 10

Q1 A rectangle is 100 mm wide and 200 mm deep. Find its I about the horizontal axis through its centroid, in units of 10^6 mm⁴.

Answer _____ × 10^6 mm⁴

Q2 For the same rectangle, find I about its base, in units of 10^6 mm⁴.

Answer _____ × 10^6 mm⁴

Q3 Find the polar moment of inertia J of an 80 mm diameter solid shaft, in units of 10^6 mm⁴.

Answer _____ × 10^6 mm⁴

Q4 An area of 5000 mm² has $\bar{I} = 2 \times 10^6$ mm⁴ about its centroidal axis. Find I about a parallel axis 40 mm away, in units of 10^6 mm⁴.

Answer _____ × 10^6 mm⁴

Q5 What is the radius of gyration of that area about the 40 mm axis?

Answer _____ mm

Q6 A solid flywheel disc has a mass of 400 kg and a diameter of 1.2 m. What is its mass moment of inertia about the shaft?

Answer _____ kg·m²

Q7 A slender rod 3 m long with a mass of 12 kg swings about one end. What is its mass moment of inertia about that end?

Answer _____ kg·m²

Q8 Of all the parallel axes, the area moment of inertia is smallest about:

(A) The base

(B) The axis through the centroid

(C) Any axis, since they're all equal

(D) The axis furthest away

Q9 An equilibrium position is stable when the potential energy V is:

(A) A maximum, $d^2V/d\theta^2 < 0$

(B) A minimum, $d^2V/d\theta^2 > 0$

(C) Zero

(D) Constant

Q10 I-beams and channel stiffeners are efficient in bending because:

(A) They are light

(B) Most of their material is far from the neutral axis, which gives a large I for their area

(C) They have a small I

(D) They are symmetric

FINAL EXAM · 20 QUESTIONS · 90 MIN

Review and final exam

NAME _____ DATE _____ SCORE ____ / 20

Q1 Two forces of 60 kN and 80 kN act at a point at 90° to each other. What is the magnitude of their resultant?

Answer _____ kN

Q2 For the forces in Q1, the equilibrant:

(A) Is 140 kN

(B) Is 100 kN, acting opposite to the resultant

(C) Is 20 kN

(D) Bisects the angle between them

Q3 A 2000 kg load hangs from two sling legs, each 45° from the vertical. What is the tension in each leg?

Answer _____ kN

Q4 A 300 N force is applied perpendicular to a 0.6 m bar. What moment does it produce about the bar's pivot?

Answer _____ N·m

Q5 A force F = (400 **i** + 300 **j**) N acts at the point (2, −1) m. What is its moment about the origin? Give a signed answer with anticlockwise positive.

Answer _____ N·m

Q6 A simply supported beam spans 10 m. It carries a uniform load of 2 kN/m along its full length and a 30 kN point load 4 m from A. What is the reaction at B?

Answer _____ kN

Q7 For the same beam, what is the reaction at A?

Answer _____ kN

Q8 A 3 m cantilever carries a 5 kN load at the free end and a uniform load of 2 kN/m along its length. What is the magnitude of the moment at the wall?

Answer _____ kN·m

Q9 A triangular truss spans 6 m. Its apex is 2 m high and carries 9 kN. What is the magnitude of the force in one sloping member?

Answer _____ kN

Q10 A simply supported beam spans 6 m. A uniform load of 4 kN/m covers only the left 3 m. What is the maximum bending moment?

Answer _____ kN·m

Q11 A 300 kg machine sits on a level deck with $\mu_s = 0.3$. What horizontal force will just start it sliding?

Answer _____ N

Q12 A 500 kg load is hauled up a 10° slope at 0.5 m/s with $\mu_k = 0.15$ and the pull parallel to the slope. What power is needed?

Answer _____ W

Q13 A rope makes 3 full turns on a bollard with $\mu = 0.25$ and holds a 25 kN load. What force must be applied on the tail end?

Answer _____ N

Q14 A screw jack will hold its load without a brake when:

(A) Its lead angle exceeds the friction angle

(B) Its friction angle exceeds the lead angle

(C) It is lubricated

(D) Its pitch is large

Q15 A 150 mm diameter shaft carries a radial load of 40 kN in a journal bearing with $\mu = 0.04$, at 200 rev/min. What power is lost to friction?

Answer _____ kW

Q16 An L-shaped area is made of a 100 × 20 mm horizontal leg (x = 0–100, y = 0–20) and a 20 × 80 mm vertical leg (x = 0–20, y = 20–100). What is $\bar{x}$?

Answer _____ mm

Q17 A ship of 5000 t has KG 7.0 m. It discharges 400 t of cargo from Kg 3.0 m. What is the new KG?

Answer _____ m

Q18 A vertical bulkhead 4 m wide has seawater 3 m deep against it. What is the resultant force?

Answer _____ kN

Q19 Find the polar moment of inertia J of a hollow shaft with outside diameter 100 mm and bore 60 mm, in units of 10^6 mm^4.

Answer _____ $\times 10^6$ mm^4

Q20 Loading heavy cargo high on deck will:

(A) Lower G and increase GM

(B) Raise G and reduce GM, so the ship is less stable

(C) Have no effect on stability

(D) Move the metacentre down to G

ANSWER KEY

Answers and worked solutions

Week 1: Units, vectors and adding forces

Q1 **(C)** A force has magnitude *and* direction. Mass, time and volume have magnitude only, so they are scalars.

Q2 **2.45 kN** W = mg = 250 × 9.81 = 2 452 N = 2.45 kN.

Q3 **(B)** The equilibrant is the single force that balances the system. It cancels the resultant exactly, so it has the same size and line and points the opposite way.

Q4 **(C)** 0.005×10^6 N = 5000 N = 5 kN.

Q5 **63.6 kN** With θ the angle between the forces, $R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)} = \sqrt{(40^2 + 30^2 + 2\cdot40\cdot30\cdot\cos 50°)} = 63.6$ kN. If you use the cosine rule on the force triangle instead, the included angle is 180° − 50° = 130°, which gives the same answer.

Q6 **21.2 °** Sine rule on the force triangle: $\sin\alpha = 30\cdot\sin(130°)/63.58$, which gives α = 21.2°. Note that sin 130° = sin 50°.

Q7 **9.83 kN** $F_x = 12\cos 35° = 9.83$ kN.

Q8 **300 N** The hypotenuse is 5, so $F_y = 500 \times 3/5 = 300$ N.

Q9 **125 N** $\Sigma F_x = 200 + 0 + 250\cos 225° = 23.22$ N. $\Sigma F_y = 0 + 300 + 250\sin 225° = 123.2$ N. $F_R = \sqrt{(23.22^2 + 123.2^2)} = 125$ N, acting at 79.3° from +x.

Q10 **(A)** The resultant lies between the difference (40 − 30 = 10 N, forces opposed) and the sum (70 N, forces in line). 50 N is the case where they are at 90°. 5 N is impossible.

Week 2: Equilibrium of a particle and free-body diagrams

Q1 **(B)** Equilibrium means ΣF = 0. A body moving at constant velocity is also in equilibrium, so 'not moving' is not the definition.

Q2 **(B)** A frictionless pulley only changes the direction of the rope. The tension is the same throughout.

Q3 **(C)** $T = W/(2\cos\theta)$, with θ measured from the vertical. θ = 30° gives 0.577W and θ = 60° gives 1.0W.

Q4 **2.83 kN** W = 500 × 9.81 = 4 905 N. $T = W/(2\cos 30°) = 2.83$ kN.

Q5 **4.9 kN** $T = 4\,905/(2\cos 60°) = 4.90$ kN. That equals the full weight of the load.

Q6 **15.6 kN** ΣF_y: $T_{AC}\sin 40° = 10$, so $T_{AC} = 15.6$ kN.

Q7 **11.9 kN** ΣF_x: $T_{AB} = T_{AC}\cos 40° = 10/\tan 40° = 11.9$ kN.

Q8 **300 N** $s = 0.450 - 0.300 = 0.150$ m. $F = ks = 2000 \times 0.150 = 300$ N.

Q9 **554 N** $|r| = \sqrt{(3^2 + 4^2 + 12^2)} = 13$ m. $F_z = 600 \times 12/13 = 554$ N.

Q10 **(B)** $\cos^2\gamma = 1 - \cos^2 60° - \cos^2 60° = 1 - 0.25 - 0.25 = 0.5$, so $\cos\gamma = 0.707$ and $\gamma = 45°$.

Week 3: Moments of a force

Q1 **(C)** Moment = force × distance, so N·m (or kN·m).

Q2 **(B)** If the line of action passes through the point, the perpendicular distance d = 0, so M = 0.

Q3 **100 N·m** $M = Fd = 250 \times 0.4 = 100$ N·m.

Q4 **86.6 N·m** $M = F\,L \sin\varphi = 250 \times 0.4 \times \sin 60° = 86.6$ N·m.

Q5 **−1 300 N·m** $M_O = xF_y - yF_x = 2(-200) - 3(300) = -400 - 900 = -1300$ N·m, which is 1300 N·m clockwise.

Q6 **38.3 kN·m** The load acts vertically, so d is the horizontal reach: $10 \cos 40° = 7.66$ m. $M = 5 \times 7.66 = 38.3$ kN·m.

Q7 **112 N·m** $\mathbf{r} \times \mathbf{F} = (2\cdot 50 - 0)\mathbf{i} - (1\cdot 50 - 0)\mathbf{j} + 0\mathbf{k} = (100\mathbf{i} - 50\mathbf{j})$ N·m. $|M| = \sqrt{(100^2 + 50^2)} = 112$ N·m.

Q8 **(B)** This is the principle of moments. Option A is transmissibility, option C is Newton's 3rd law, and option D is a property of couples.

Q9 **−80 N·m** $\Sigma M = +200(0.5) - 150(1.2) = 100 - 180 = -80$ N·m, which is 80 N·m clockwise.

Q10 **(B)** d is the shortest distance from the point to the force's line of action, not to where the force happens to be drawn.

Week 4: Couples, equivalent systems and distributed loads

Q1 **(B)** The moment of a couple, F·d, does not depend on the point you take moments about.

Q2 **37.5 N·m** $M = Fd = 150 \times 0.25 = 37.5$ N·m.

Q3 **9.6 kN·m** $M = Fd = 12 \times 0.8 = 9.6$ kN·m.

Q4 **40 kN** $F = wL = 5 \times 8 = 40$ kN, acting at 4 m.

Q5 **27 kN** $F = \tfrac{1}{2} \times 9 \times 6 = 27$ kN.

Q6 **4 m** The centroid of a triangle is ⅔ of the way from the zero end: $\tfrac{2}{3} \times 6 = 4$ m from A, which is 2 m from B.

Q7 **3.5 m** $\bar{x} = (2\cdot 1 + 5\cdot 3 + 3\cdot 6)/10 = 35/10 = 3.50$ m.

Q8 **(B)** This is transmissibility.

Q9 **−15 N·m** $\Sigma M = +40 - 25 - (100 \times 0.3) = 40 - 25 - 30 = -15$ N·m, which is 15 N·m clockwise.

Q10 **2.33 m** Rectangle: $2 \times 4 = 8$ kN at 2 m. Triangle: $\frac{1}{2} \times 4 \times 4 = 8$ kN at $\frac{2}{3} \times 4 = 2.667$ m. $\bar{x} = (8 \cdot 2 + 8 \cdot 2.667)/16 = 2.33$ m. The resultant is 16.0 kN.

Week 5: Equilibrium of a rigid body

Q1 **(C)** Two force components plus a moment: A_x, A_y and M_A.

Q2 **(A)** A roller only stops motion into or away from the surface.

Q3 **12 kN** ΣM_B: $R_A(5) = 20(3)$, so $R_A = 12$ kN. Then $R_B = 8$ kN.

Q4 **21 kN·m** $M_A = 4(3) + (2 \times 3)(1.5) = 12 + 9 = 21$ kN·m.

Q5 **10 kN** $A_y = 4 + 2 \times 3 = 10$ kN.

Q6 **−3 kN** ΣM_B: $-R_A(4) - 6(2) = 0$, so $R_A = -3$ kN. That means A has to be held *down* with 3 kN. $R_B = 9$ kN.

Q7 **(B)** Only collinear, equal and opposite forces can give both $\Sigma F = 0$ and $\Sigma M = 0$.

Q8 **800 N** Lever: $200 \times (1.5 - 0.3) = F \times 0.3$, so $F = 240/0.3 = 800$ N.

Q9 **10 kN** ΣM_A: $(T \sin 30°)(4) = 10(2)$, so $T = 10.0$ kN.

Q10 **8.66 kN** ΣF_x: $A_x = T \cos 30° = 8.66$ kN.

Week 6: Trusses, frames and machines

Q1 **(B)** These assumptions make every member a two-force member.

Q2 **(A)** $\Sigma F = 0$ in two directions with two non-parallel forces can only be met if both are zero.

Q3 **(C)** There are three equilibrium equations for the free section.

Q4 **10 kN** Reactions are 6 kN each. The member runs 4 m, rises 3 m and is 5 m long. $F = 6 \times 5/3 = 10$ kN.

Q5 **(B)** They push up against the apex load, so they are in compression. The bottom tie is in tension.

Q6 **8 kN** $F_{tie} = 10 \times 4/5 = 8$ kN (T).

Q7 **60 kN** $F \times 2 = 30(6) - 20(3) = 120$, so $F = 60.0$ kN. It is in tension.

Q8 **400 N** Moments about the pivot: $100 \times 120 = F \times 30$, so $F = 400$ N. MA = 4.

Q9 **100 N** ΣM_B: $50 \times 200 = F \times 100$, so $F = 100$ N.

Q10 **(B)** This is Newton's 3rd law: action and reaction.

Week 7: Internal forces: shear force and bending moment diagrams

Q1 **27 kN·m** $M_{max} = PL/4 = 18 \times 6/4 = 27$ kN·m.

Q2 **37.5 kN·m** $M_{max} = wL^2/8 = 3 \times 100/8 = 37.5$ kN·m.

Q3 **20 kN·m** $M_{max} = PL = 8 \times 2.5 = 20$ kN·m, at the wall. It is hogging, so negative in Hibbeler's convention.

Q4 **6 kN** $V_{max} = wL = 1.5 \times 4 = 6$ kN, at the wall.

Q5 **(B)** dM/dx = V, so M has a turning point where V = 0.

Q6 **(B)** dM/dx = V.

Q7 **(B)** Constant w makes V linear in x and M quadratic in x.

Q8 **36 kN·m** $R_A = 24 \times 6/8 = 18.0$ kN. M_{max} is under the load: $18.0 \times 2 = 36.0$ kN·m. Or use $Pa(L - a)/L = 24 \cdot 2 \cdot 6/8$.

Q9 **18 kN·m** $R_B = 24 - 18.0 = 6.00$ kN. Working from the right: $M = 6.00 \times 3 = 18.0$ kN·m.

Q10 **(B)** Excess weight amidships bends the hull concave-up. That is sagging, with the deck in compression and the keel in tension.

Week 8: Dry friction and wedges

Q1 **(B)** Static friction only reaches $\mu_s N$ when slipping is about to happen.

Q2 **200 N** $F = \mu_s N = 0.4 \times 500 = 200$ N.

Q3 **0.404** $\mu_s = \tan 22° = 0.404$.

Q4 **172 N** $P = \mu_s mg = 0.35 \times 50 \times 9.81 = 172$ N.

Q5 **443 N** $P = W(\sin 15° + \mu \cos 15°) = 981(0.2588 + 0.1932) = 443$ N.

Q6 **2.22 kJ** $U = Ps = 443.4 \times 5 = 2\,217$ J $= 2.22$ kJ.

Q7 **(B)** A heap or a block on a slope stays put up to $\theta = \varphi_s$.

Q8 **278 N** $P = W(\sin 30° - \mu \cos 30°) = 981(0.5 - 0.2165) = 278$ N. Friction helps here by acting up the slope.

Q9 **(A)** Whichever condition needs the smaller push happens first.

Q10 **200 N** Tip: $P = W(b/2)/h = 800 \times 0.3/1.2 = 200$ N. Slide: $P = 0.4 \times 800 = 320$ N. Tipping needs less, so it tips at 200 N.

Week 9: Screws, belts, bearings and rolling resistance

Q1 **1 097 N** $T_2 = 500\, e^{0.25\pi} = 500 \times 2.193 = 1\,097$ N.

Q2 **2.44 turns** $\beta = \ln(15000/150)/0.3 = 15.4$ rad. Turns $= \beta/2\pi = 2.44$. In practice you'd take 3.

Q3 **389 N** $\beta = 160° = 2.793$ rad. $T_1 = 900 / e^{0.3 \times 2.793} = 389$ N.

Q4 **5.11 kW** $P = (T_2 - T_1)v = (510.6) \times 10 = 5\,106$ W $= 5.11$ kW.

Q5 **86.3 N·m** $\theta = \tan^{-1}(8/(\pi \cdot 40)) = 3.64°$. $\varphi = \tan^{-1} 0.15 = 8.53°$. $M = 20\,000 \times 0.020 \times \tan(12.2°) = 86.3$ N·m.

Q6 **(B)** If $\varphi_s > \theta$, friction alone holds the load. In Q5 $\varphi = 8.53°$ is greater than $\theta = 3.64°$, so that jack is self-locking.

Q7 **2.36 kW** $M \approx \mu P r = 0.05 \times 30\,000 \times 0.05 = 75.0$ N·m. $\omega = 2\pi \times 300/60 = 31.42$ rad/s. Power $= M\omega = 2\,356$ W $= 2.36$ kW.

Q8 **62.2 N·m** $M = \frac{2}{3} \times 0.08 \times 10\,000 \times (0.1^3 - 0.05^3)/(0.1^2 - 0.05^2) = 62.2$ N·m.

Q9 **65.4 N** $P \approx Wa/r = (2000 \times 9.81)(0.0005)/0.15 = 65.4$ N.

Q10 **(B)** β must be in radians, and T_2 is on the side being pulled toward the slip.

Week 10: Centre of gravity, centroids and fluid pressure

Q1 **143 mm** Web: $A = 3600$ mm², $\tilde{y} = 90$. Flange: $A = 4000$ mm², $\tilde{y} = 190$. $\bar{y} = (3600 \cdot 90 + 4000 \cdot 190)/7600 = 143$ mm.

Q2 **142 mm** Plate: 60 000 mm² at 150. Hole: −7 854 mm² at 200. $\bar{x} = (60\,000 \cdot 150 - 7\,854 \cdot 200)/(60\,000 - 7\,854) = 142$ mm. The hole shifts the centroid away from itself.

Q3 **(B)** Triangle: h/3 from the base, which is 2h/3 from the apex.

Q4 **5.5 m** $KG = (2000 \cdot 6 + 500 \cdot 2 + 300 \cdot 8)/2800 = 15\,400/2800 = 5.50$ m.

Q5 **60.3 kPa** $p = \rho g h = 1025 \times 9.81 \times 6 = 60\,332$ Pa $= 60.3$ kPa.

Q6 **19.6 kN** $F = \rho g \bar{h} A = 1000 \times 9.81 \times 1 \times 2 = 19.6$ kN.

Q7 **1.33 m** For a rectangle that reaches the surface, $h_{cp} = \frac{2}{3} \times 2 = 1.333$ m.

Q8 **32.8 kN** $\bar{h} = 3 + 0.4 = 3.4$ m. $A = 0.96$ m². $F = 1025 \times 9.81 \times 3.4 \times 0.96 = 32.8$ kN.

Q9 **1.58 litres** $V = 2\pi \bar{r} A = 2\pi(0.2)(\pi \times 0.02^2) = 0.001579$ m³ $= 1.58$ L.

Q10 **(B)** Pressure increases with depth, so the resultant acts below the centroid. The gap is $I_G/(\bar{h}A)$.

Week 11: Moments of inertia and stability

Q1 **66.7 × 10⁶ mm⁴** $\bar{I} = bh^3/12 = 100 \times 200^3/12 = 66.7 \times 10^6$ mm⁴.

Q2 **267 × 10⁶ mm⁴** $I = bh^3/3 = 267 \times 10^6$ mm⁴. Check: $\bar{I} + Ad^2 = 66.67 + 20\,000 \times 100^2 / 10^6 = 266.7$ ✓

Q3 **4.02×10^6 mm⁴** $J = \pi d^4/32 = \pi \times 80^4/32 = 4.02 \times 10^6$ mm⁴.

Q4 **10×10^6 mm⁴** $I = \bar{I} + Ad^2 = 2 \times 10^6 + 5000 \times 40^2 = 10 \times 10^6$ mm⁴.

Q5 **44.7 mm** $k = \sqrt{(I/A)} = \sqrt{(10 \times 10^6/5000)} = 44.7$ mm.

Q6 **72 kg·m²** $I = \frac{1}{2}mr^2 = \frac{1}{2} \times 400 \times 0.6^2 = 72$ kg·m².

Q7 **36 kg·m²** $I = mL^2/3 = 12 \times 9/3 = 36$ kg·m². Or $I_G + md^2 = 12 \cdot 9/12 + 12 \cdot 1.5^2 = 9 + 27 = 36$ ✓

Q8 **(B)** $I = \bar{I} + Ad^2$ is smallest when $d = 0$.

Q9 **(B)** At a minimum of V, any small displacement raises V, and the body returns.

Q10 **(B)** The Ad^2 terms dominate, so putting the area in the flanges is efficient.

Final exam: Review and final exam

Q1 **100 kN** $R = \sqrt{(60^2 + 80^2)} = 100$ kN.

Q2 **(B)** The equilibrant is equal and opposite to the resultant.

Q3 **13.9 kN** $T = W/(2 \cos 45°) = 19.62/(1.414) = 13.9$ kN.

Q4 **180 N·m** $M = 300 \times 0.6 = 180$ N·m.

Q5 **1 000 N·m** $M = xF_y - yF_x = 2(300) - (-1)(400) = 1000$ N·m, anticlockwise.

Q6 **22 kN** ΣM_A: $10R_B = 20(5) + 30(4) = 220$, so $R_B = 22.0$ kN.

Q7 **28 kN** $R_A = 20 + 30 - 22.0 = 28.0$ kN.

Q8 **24 kN·m** $M = 5 \times 3 + 2 \times 3 \times 1.5 = 15 + 9 = 24$ kN·m.

Q9 **8.11 kN** Reactions are 4.5 kN each. The member is $\sqrt{(3^2 + 2^2)} = 3.606$ m long. $F = 4.5 \times 3.606/2 = 8.11$ kN (C).

Q10 **10.1 kN·m** $R_B = 12 \times 1.5/6 = 3$ kN, so $R_A = 9$ kN. $V = 9 - 4x = 0$ at $x = 2.25$ m. $M_{max} = 9(2.25) - 2(2.25^2) = 10.1$ kN·m.

Q11 **883 N** $F = 0.3 \times 300 \times 9.81 = 883$ N.

Q12 **788 W** $P_{force} = 4905(\sin 10° + 0.15 \cos 10°) = 1\,576$ N. Power $= Fv = 1\,576 \times 0.5 = 788$ W.

Q13 **225 N** $\beta = 3 \times 2\pi = 18.85$ rad. $T_1 = 25\,000 / e^{0.25 \times 18.85} = 225$ N.

Q14 **(B)** Self-locking needs $\varphi_s > \theta$.

Q15 **2.51 kW** $M = 0.04 \times 40\,000 \times 0.075 = 120$ N·m. $\omega = 20.94$ rad/s. $P = 2\,513$ W $= 2.51$ kW.

Q16 **32.2 mm** $\bar{x} = (2000 \times 50 + 1600 \times 10)/3600 = 32.2$ mm.

Q17 **7.35 m** KG = (5000 × 7 − 400 × 3)/4600 = 7.35 m. Taking weight off low down raises G.

Q18 **181 kN** F = 1025 × 9.81 × 1.5 × 12 = 181 kN, acting 2 m below the surface.

Q19 **8.55×10^6 mm⁴** $J = \pi(100^4 - 60^4)/32 = 8.55 \times 10^6$ mm⁴.

Q20 **(B)** Raising KG reduces GM = KM − KG, which reduces initial stability.